

Differential Expression of SRY-Related HMG-Box Transcription Factor 2, Oligodendrocyte Lineage Transcription Factor 2, and Zinc Finger E-Box Binding Homeobox 1 in Serum-Derived Extracellular Vesicles: Implications for Mithramycin Sensitivity and Targeted Therapy in High-Grade Glioma

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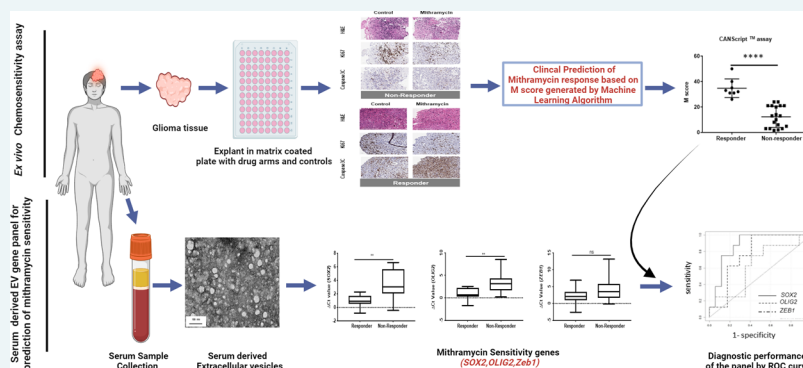
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ABSTRACT: Glioblastoma multiforme (GBM) is the most aggressive type of glioma and is often resistant to traditional therapies. Evidence suggests that glioma stem cells (GSCs) contribute to this resistance. Mithramycin (Mit-A) targets GSCs and exhibits antitumor activity in GBM by affecting transcriptional targets such as SRY-related HMG-box transcription factor 2 (SOX2), oligodendrocyte lineage transcription factor 2 (OLIG2), and zinc finger E-box binding homeobox 1 (ZEB1). However, its clinical use has been limited by toxicity. This study explored the diagnostic potential of serum extracellular vesicles (EVs) to identify Mit-A responders. Serum EVs were isolated from 70 glioma patients, and targeted gene expression was analyzed using qRT-PCR. Using chemosensitivity assay, we identified 8 Mit-A responders and 17 nonresponders among 25 glioma patients. The M-score showed a significant correlation ($p = 0.045$) with isocitrate dehydrogenase 1 mutation but not other clinical variables. The genes SOX2 ($p = 0.005$), OLIG2 ($p = 0.003$), and ZEB1 ($p = 0.0281$) were found to be upregulated in the responder EVs. SOX2 had the highest diagnostic potential (AUC = 0.875), followed by OLIG2 (AUC = 0.772) and ZEB1 (AUC = 0.632). The combined gene panel showed significant diagnostic efficacy (AUC = 0.956) through logistic regression analysis. The gene panel was further validated in the serum EVs of 45 glioma patients. These findings highlight the potential of Mit-A as a targeted therapy for high-grade glioma based on differential gene expression in serum EVs. The gene panel could serve as a diagnostic tool to predict Mit-A sensitivity, offering a promising approach for personalized treatment strategies and emphasizing the role of GSCs in therapeutic resistance.

KEYWORDS: mithramycin, glioblastoma, CANScript™ assay and extracellular vesicles

Gliomas are the most prevalent adult brain tumors within the central nervous system. Glioblastoma multiforme (GBM), the most aggressive subtype of gliomas, has a dismal prognosis, with a median survival time of approximately 12–15 months following standard treatments, including maximal surgical resection, radiotherapy, and adjuvant chemotherapy with temozolomide.^{1,2} Unfortunately, due to the highly heterogeneous nature of GBM and the development of resistance to

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Table 1. Clinico Pathological Characteristics of 70 Glioma Patients

characteristics	grade I (n= 4)	grade II (n= 14)	grade III (n= 13)	grade IV (n= 39)
frequency	5.71%	20%	18.58%	55.71%
	gender			
male	2 (50%)	5 (35.72%)	8(61.54%)	26(66.67%)
female	2(50%)	9 (64.28%)	5(38.46%)	13(33.33%)

conventional therapies, relapses are frequent, and no defined standard of care exists for these patients.^{3,4}

GBM is highly heterogeneous, comprising multiple cellular subpopulations, notably, glioma stem cells (GSCs). These cells possess the hallmark characteristics of neural stem cells, such as self-renewing and differentiating into multiple cell types within the tumor.⁵ GSCs have been implicated in initiating and maintaining glioblastoma and are resistant to chemotherapy, radiation, and targeted drugs.⁶ This resistance significantly contributes to tumor recurrence and treatment failure, highlighting GSCs as promising therapeutic targets in glioma management. A group of transcription factors, namely SOX2 (SRY-related HMG-box transcription factor 2), OLIG2 (oligodendrocyte lineage transcription factor 2), and ZEB1 (zinc finger E-box binding homeobox 1), have been associated with GSCs.^{7–10} High expression levels of these factors, particularly SOX2, have been linked with more aggressive tumors and poorer prognosis in GBM patients.^{11–13} SOX2 is a key transcription factor of the SOX family and a marker of GSC.¹⁴ Evidence suggests that SOX2, independently of receptor tyrosine kinases (RTK), governs glioma-initiating cells through its target genes, OLIG2 and ZEB1.⁷ Therefore, drugs inhibiting these transcription factors may offer effective therapeutic options for glioma treatment.

Mithramycin (Mit-A), an antineoplastic antibiotic known to inhibit RNA synthesis and produced by *Streptomyces plicatus*, is one such promising compound.¹⁵ Mit-A has been shown to have antitumor effects by targeting cancer stem cells in various cancers.^{16–19} Mit-A has demonstrated significant inhibition of GBM invasiveness by targeting the GSC transcription factors SOX2, OLIG2, and ZEB1 in glioma.⁷ Additionally, Mit-A treatment reduced cell migration in gliomas by altering MMP-2 and 9, VEGF, and RON kinase.²⁰ In a screen of 147 FDA-approved oncology drugs, Mit-A and six other drugs demonstrated promising antiproliferative activity in GSCs of high-grade isocitrate dehydrogenase (IDH) mutant glioma.²¹ Another study showed that Mit-A inhibits invasiveness in high-grade gliomas.²² Mit-A causes severe side effects such as trasaminitis²³ and hepatotoxicity.²⁴ Mit-A derivatives have potent antitumor activity, and decreased toxicities will make the compound an attractive drug candidate for glioma treatment due to its ability to cross the blood-brain barrier to modulate the transcriptional targets of GSCs, including SOX2, OLIG2, and ZEB1.^{7,25} By directly targeting the transcriptional machinery supporting GSCs, Mit-A offers a unique opportunity to combat tumor heterogeneity and drug resistance, potentially improving the prognosis for glioma patients. However, the precise impact of Mit-A on GSC behavior and glioma progression requires further research.

The clinical application of Mit-A in glioma therapy is challenged by predicting the patient response. To better understand the potential role of Mit-A in glioma treatment, we employed CANScrip, a human *ex vivo* chemosensitivity assay. This assay preserves tumor heterogeneity and the micro-environment, enabling a more accurate prediction of a patient's

response to a specific drug.²⁶ This assay has been used as a drug sensitivity tool for various cancers, including GBM.^{27–29}

Extracellular vesicles (EVs), the small membrane-enclosed particles released by tumor cells, have emerged as key players in modulating the tumor microenvironment. Their influence extends to the development of tumor heterogeneity, a critical factor in cancer progression and treatment resistance. EVs reflect the molecular characteristics of their cells of origin, acting as highly specialized entities of communication.³⁰ They shuttle cargo rich in surface markers, signaling molecules, oncogenic proteins, and nucleic acids, which can be horizontally transferred to stromal target cells. This transfer can fundamentally alter the tumor microenvironment, promoting drug resistance as well as tumor growth, invasion, and metastasis.³¹

The dynamic and transient contributions of EVs to tumor heterogeneity are mediated through their diverse molecular cargo. Deciphering this cargo is essential to understanding the complexities of tumor heterogeneity and the development of precise, patient-specific therapies. The use of EVs in cancer diagnostics and therapeutics has been increasingly recognized, with their cargo offering insights into cancer prognosis and aiding in therapy monitoring for conditions such as glioma.³² EV-based molecular biomarkers represent a promising strategy for accurate patient stratification and prognostic staging, an area that requires urgent attention.³³

In light of these advances, our research aims to establish a sensitive and minimally invasive liquid biopsy-based diagnostic tool to assess Mit-A sensitivity in glioma. Although the potential of serum EV cargo as predictive markers for therapeutic response is recognized,³⁴ the specific role and reliability of these EVs in determining Mit-A responsiveness necessitate further exploration. Our focus in this work is to validate the utility of EVs in this context, ensuring that this novel approach to liquid biopsy can effectively guide personalized treatment strategies.

Addressing these limitations forms the primary motivation for our study. We aim to provide a more comprehensive understanding of the Mit-A therapeutic potential by studying its effects on GSC transcription factors. Concurrently, we explore the potential of serum-derived EVs as predictive markers for the Mit-A response. We hypothesize that Mit-A will demonstrate effective action against GSCs and that the SOX2, OLIG2, and ZEB1 expression profiles in EVs can predict Mit-A response. By addressing these crucial gaps in the field, our work hopes to enable a more personalized and potentially more effective approach to glioma treatment.

RESULTS

Clinicopathological Profile of Glioma Patients. This study included 70 pathologically confirmed glioma patients and 10 healthy age- and gender-matched individuals. All patient-derived tumor tissue samples were categorized based on the World Health Organization (WHO) grading criteria, and investigators remained blinded to the tumor grades and

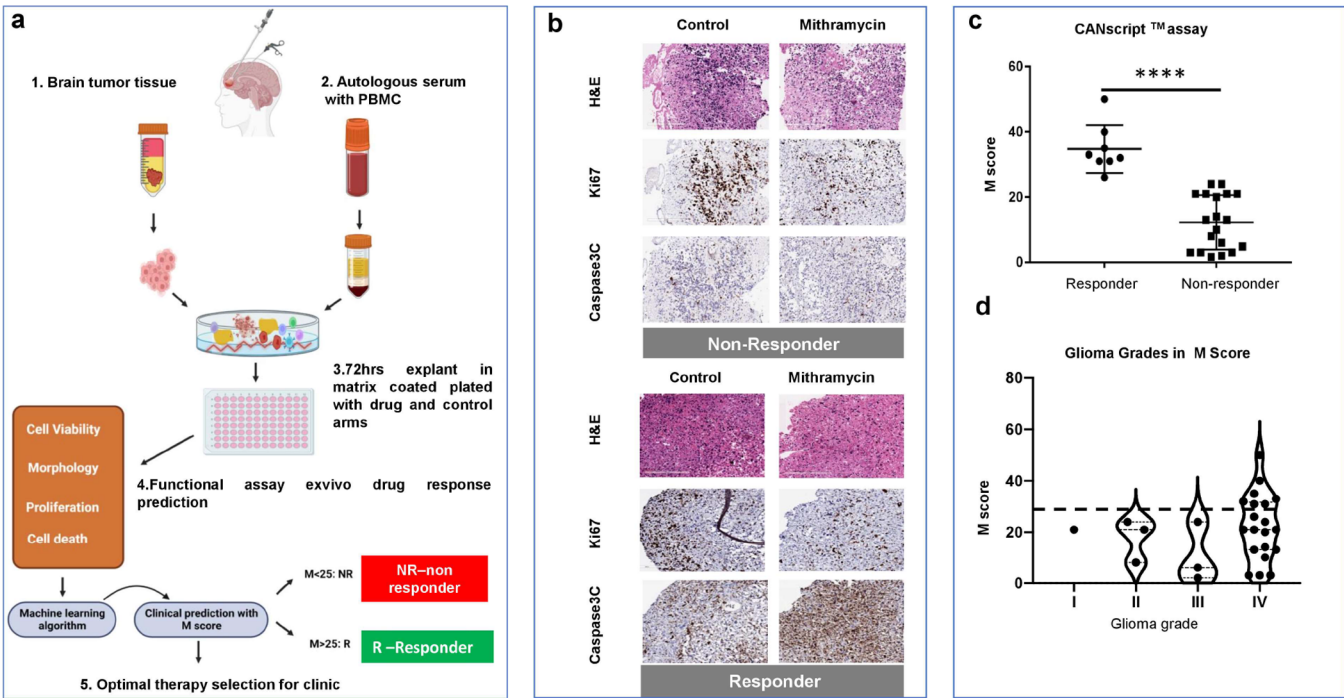


Figure 1. Establishment of the M-Score in glioma tissues. (a) Schematic of CANScript ex vivo testing, detailing the flow from the glioma patient tumor slice culture to the generation of a predictive score (M-Score). (b) Representative histological and immunohistochemical staining of glioma explants from responder and nonresponder samples. The panels display tumor content (hematoxylin and eosin staining), proliferation (Ki-67 marker), and viability (Caspase 3C) upon treatment with Mit-A. The scale bar represents 200 μ m. (c) Graphical representation of the number of glioma cases responding to Mit-A treatment, categorized into responder and nonresponder populations. (d) Chart depicting the M-Score distribution among glioma samples, stratified grade-wise. P -value < 0.05 is considered to be significant, (* p < 0.05; ** p < 0.01, and *** p < 0.001).

patient outcomes. A summary of the clinicopathological features of the glioma patients is provided in Table 1. The patient cohort had a median age of 48.5 years, ranging from 18 to 84 years. A breakdown by glioma grade revealed the following distribution: grade I accounted for 5.71% of the patient population with a median age of 54 years, grade II constituted 20% with a median age of 40.5 years, grade III represented 18.58% with a median age of 38 years, and grade IV, the most prevalent grade, made up 55.71% of the patients with a median age of 54 years. Sample assignment for the study was conducted randomly to minimize selection bias.

Mit-A Sensitivity in Glioma Patients Based on CANScript™ Assay and Clinical Variables. We examined the sensitivity of Mit-A across histologically confirmed glioma samples of various grades ($n = 25$) using ex vivo assays. In these assays, patient-derived tumor explants were treated with Mit-A at 100 nM/ml (Figure 1a). The chemosensitivity to Mit-A was gauged by evaluating phenotypic attributes such as changes in tumor morphology, viability (H&E staining), proliferation (Ki67), and the induction of apoptosis (Caspase 3C). The functional outcomes from this assay were transformed into a single score known as "M-Score," designed to predict the patient's clinical response to Mit-A. Notably, the M-score revealed differential profiles in responders and nonresponders, mainly, loss of viable tumor cells and increased expression of Caspase 3C were observed in the responder population (Figure 1b). The M-Score analysis revealed that 8 of the 25 glioma patients responded positively to Mit-A treatment, while 17 were nonresponsive (Figure 1c). The M-scores from the glioma samples were categorized based on tumor grades: grade 1 (1), grade 2 (3), grade 3 (3), and grade 4 (18) (Figure 1d). In examining the relationship with various

clinical variables, we found that age, gender, tumor grade, and ATRX gene expression did not significantly correlate with Mit-A sensitivity or the M-Score. However, all samples in the predicted responder population expressed wild-type IDH1 and correlated considerably with M-Score ($p = 0.045$) (Table 2).

Serum EV Characterization in Glioma Patients. The purity of the isolated serum EVs was verified by biophysical,

Table 2. Clinicopathological Characteristics of CANScript™ Glioma Study Patients

characteristics	responders (M-score > 25)	nonresponders (M-score < 25)	P-value
age			1.00
>49 years	4(33.3%)	8(66.7%)	
<49 years	4(30.8%)	9(69.2%)	
gender			0.667
male	4(50%)	11(64.7%)	
female	4(50%)	6(35.3%)	
grade			0.206
I	0/8 (0%)	1/17 (5.88%)	
II	0/8(0%)	3/17(17.6%)	
III	0/8(0%)	3/17(17.6%)	
IV	8/8(100%)	10/17 (58.8%)	
IDH mutation			0.045*
wild-type	5 (100%)	6 (42.9%)	
IDH mutant	0 (0%)	8(57.1%)	
total 19/25	5 (100%)	14(100%)	
[ATRX total]			0.519
expression	4(100%)	9 (69.2%)	
loss	0(0%)	4 (30.8%)	
total	4 (100%)	13(100%)	

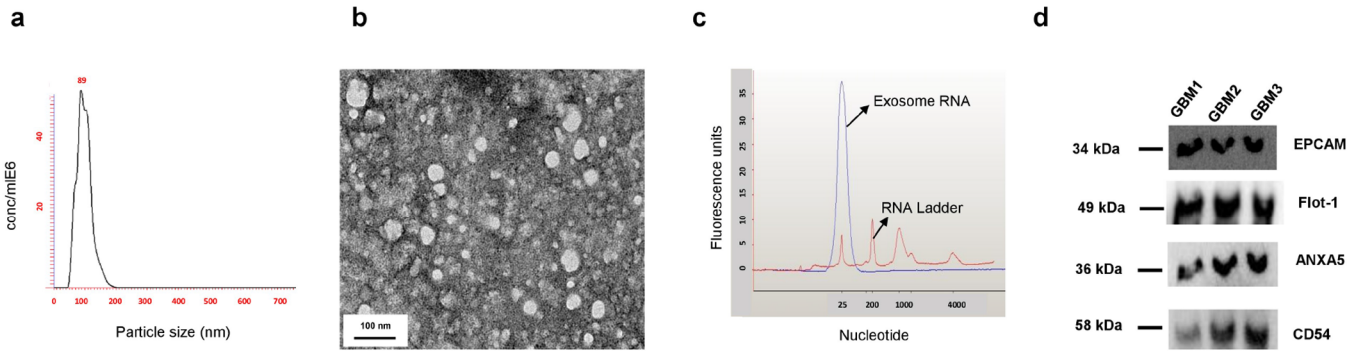


Figure 2. Characterization of GBM serum EVs. This figure illustrates the characterization of EVs from the serum of GBM patients, providing insights into their size, structure, and molecular markers. (a) Representative NTA showing isolated EV size distribution profiles and concentrations. (b) A TEM image of EVs, displaying their morphology at a nanoscale level (scale Bar: 100 nm) (c) An electropherogram generated using the Agilent 2100 Bioanalyzer, representing the RNA profile of serum EVs. (d) Western blot analysis of specific GBM serum EV markers, including EpCAM (epithelial cell adhesion molecule), flotillin-1, annexin-5, and CD54.

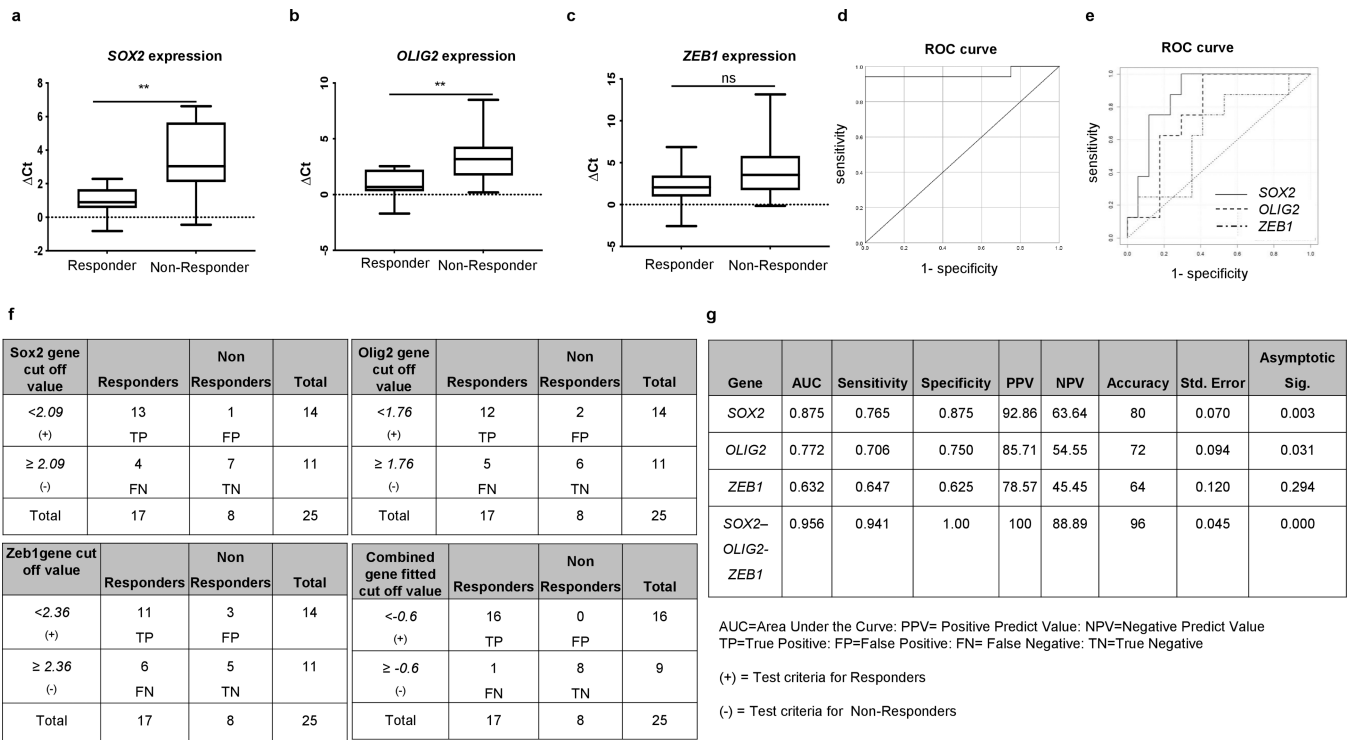


Figure 3. This figure illustrates the process of establishing a companion diagnostic panel for Mit-A responsive genes and evaluating their diagnostic accuracy. EVs were isolated from serum samples, and the expression levels of Mit-A responsive genes were quantified in both responder and nonresponder groups. (a–c) Panels display the expression levels (delta Ct values) of SOX2, OLIG2, and ZEB1 genes in serum EVs from Mit-A responders ($n = 8$) and nonresponders ($n = 17$). (d) Combined ROC curve for SOX2, OLIG2, and ZEB1, showing the collective diagnostic accuracy of the gene panel. (e) Individual ROC curves for SOX2, OLIG2, and ZEB1 illustrate their diagnostic potential. (f) 2×2 diagnostic matrix for SOX2, OLIG2, and ZEB1 defining the Mit-A response. (g) Table summarizing the characteristics of the ROC curves for each gene, providing a concise overview of the findings. A p -value of less than 0.05 is considered statistically significant. (* $p < 0.05$; ** $p < 0.01$; and *** $p < 0.001$).

biochemical, and surface membrane profiling. The size of the isolated EVs was determined through nanoparticle tracking analysis (NTA), which revealed their diameter to be within the 30–200 nm range, as demonstrated in Figure 2a. The morphology of the isolated EVs was then corroborated through transmission electron microscopy (TEM) analysis (Figure 2b). This examination confirmed that the serum-derived EVs were intact, double-membrane structures exhibiting a round shape and a size range of 30–150 nm. The size and quality of the RNA associated with the EVs were assessed using a bioanalyser. The EV-associated RNA ranged

between 20 and 180 nucleotides in length (Figure 2c). An identifiable peak in the 5S region suggested an enrichment of EV RNA against the 25-nucleotide region of the internal standard. The absence of 18S and 28S peaks in the EV RNA spectrum substantiated its purity, verifying the absence of cellular RNA contamination. Further verification of the successful EV isolation was confirmed by detecting EVs enriched with proteins annexin-5, flotillin-1, EpCAM, and CD54 (Figure 2d) by Western blot analysis. These results confirm the purity of our EV preparation, considering their size, shape, and molecular and protein profiles.

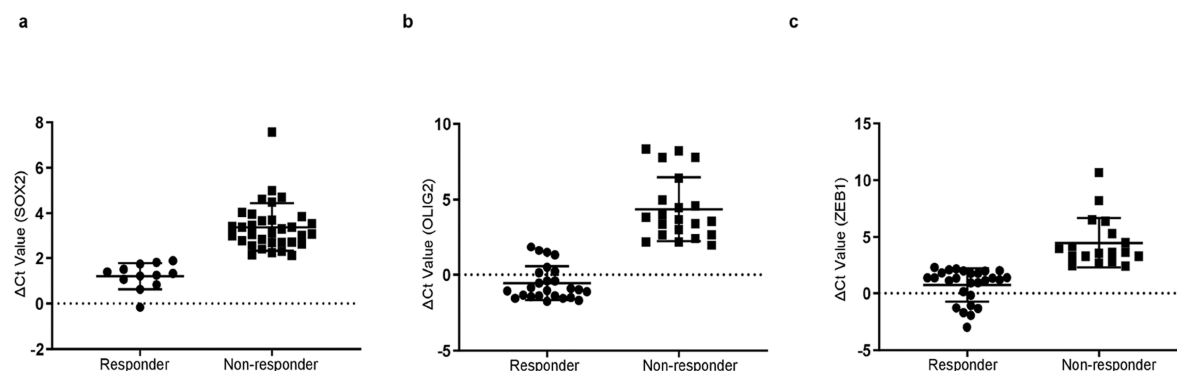


Figure 4. Stratification of glioma patients based on the Mit-A companion diagnostic panel. This figure illustrates the stratification of glioma patients ($n = 45$) into Mit-A responders and nonresponders. The stratification was performed using ROC-derived cutoff values of individual gene expression in serum EVs (a) Stratification based on the cutoff value of SOX2 (i.e., 2.096). (b) Stratification based on the cutoff value of OLIG2 (i.e., 1.76). (c): Stratification based on the cutoff value of ZEB1 (i.e., 2.36).

Expression of SOX2, OLIG2, and ZEB1 in Glioma Tissues and Paired Serum EVs. To understand the effects of Mit-A, we analyzed the expression of its critical target genes: SOX2, OLIG2, and ZEB1. This analysis was conducted in glioma tissues and serum-derived EVs of grades III and IV using semiquantitative PCR and RT-qPCR methods. The employed primer sets produced the expected amplicon sizes of SOX2 (150 bp), OLIG2 (72 bp), ZEB1 (117 bp), and GAPDH (130 bp) used as an internal control (Figure S1a,b). A no-template control was used as the negative control.

Further, we quantified the expression levels of these genes across all grades of glioma tissues and serum EVs. Our findings revealed a statistically significant upregulation of SOX2 ($p < 0.0001$), OLIG2 ($P < 0.0028$), and ZEB1 ($P < 0.003$) in glioma tissues when compared to healthy controls (Figure S2a–c). Similarly, the serum EVs showed significantly elevated expression of SOX2 ($p < 0.0001$) and OLIG2 ($P < 0.0001$) (Figure S2d–f). However, ZEB1 expression in glioma serum EVs was not found to be significantly different from the healthy controls ($P < 0.225$).

Differential Expression of SOX2, OLIG2, and ZEB1 in Mit-A Responders and Nonresponders. Given the lack of discernible associations between ex vivo response prediction and clinical correlates, excluding IDH mutational status, we focused on profiling the critical gene expressions associated with Mit-A. In this context, we quantified the SOX2, OLIG2, and ZEB1 transcript levels in Mit-A responders and non-responders of serum-derived EVs. Our analysis revealed a notable disparity in the expression of SOX2 and OLIG2 between these two groups. We observed a statistically significant increase in the expression levels of SOX2 ($P = 0.005$) and OLIG2 ($P = 0.003$) in patients whose tumors responded to Mit-A treatment compared to the nonresponder group. However, this trend did not extend to the ZEB1 expression. Our analysis found no significant variation in ZEB1 expression between the two groups ($P = 0.281$) (Figure 3a–c).

Evaluating the Diagnostic Performance of SOX2, OLIG2, and ZEB1 in Predicting Mit-A Response. The differential expression levels of SOX2, OLIG2, and ZEB1 in serum-derived EVs from the Mit-A responders versus non-responders group indicated the potential utility of these transcripts as biomarkers for predicting Mit-A response in glioma. We further performed receiver operating characteristic (ROC) curve analyses to assess these mRNAs' applicability as diagnostic biomarkers for Mit-A response in glioma. The ROC

analysis in Figure 3d–g reveals that SOX2 exhibited the highest area under the curve (AUC) value of 0.875 [95% confidence interval (CI): 0.738–1.00, $P < 0.003$], indicating a strong predictive capacity (87.5%) to differentiate Mit-A responders from nonresponders. The optimal cutoff value of 2.096 yielded the following performance metrics: sensitivity (76.5%), specificity (87.5%), positive predictive value (PPV) (92.86%), negative predictive value (NPV) (63.64%), and accuracy (80%). Patients with SOX2 expression levels greater than 2.096 were categorized as nonresponders, while those with levels below 2.096 were considered responders to Mit-A. Following SOX2, OLIG2 exhibited the second-highest AUC of 0.772 (95% CI: 0.588–0.956, $P < 0.031$), suggesting a predictive power of 77% in distinguishing Mit-A response in glioma patients. Nonresponders were defined as individuals with OLIG2 expression levels exceeding 1.76, while responders had levels less than 1.76. At this optimal cutoff value, the performance metrics were as follows: sensitivity (70%), specificity (75%), PPV (85.71%), NPV (54.55%), and accuracy (72%). ZEB1, on the other hand, displayed a lower AUC of 0.632 (95% CI, 0.397–0.867, $P < 0.294$), providing a less robust differentiation (63%) in Mit-A response. At an optimal cutoff value of 2.36, the performance metrics for ZEB1 were as follows: sensitivity (64.7%), specificity (62.5%), PPV (78.57%), NPV (45.45%), and accuracy (64%).

To enhance diagnostic precision, we integrated these three biomarkers into a logistic regression model. The regression fitted values were then used for ROC analysis, resulting in significantly improved diagnostic efficacy (AUC = 0.956, 95% CI: 0.868–1.000; $P < 0.000$). This integrated approach achieved a sensitivity of (94.1%), specificity of (100%), PPV of (100%), NPV of (88.89%), and accuracy of (96%) at an optimal cutoff value of -0.6 .

Patient Stratification in Glioma Using a Three-Gene Companion Diagnostic Panel. The diagnostic potential of the gene panel (SOX2, OLIG2, and ZEB1) developed through ROC prompted us to validate its predictive power in an independent validation cohort of 45 glioma patients to predict the Mit-A response, independent of an ex vivo assay. To validate, we quantified the mRNA expression levels of SOX2, OLIG2, and ZEB1 in serum-derived EVs of 45 glioma patients of different tumor grades (grade I–IV). We stratified them as Mit-A responders and nonresponders based on the individual optimal gene expression cutoff values derived from the ROC analysis of the discovery cohort. ROC curve was derived the

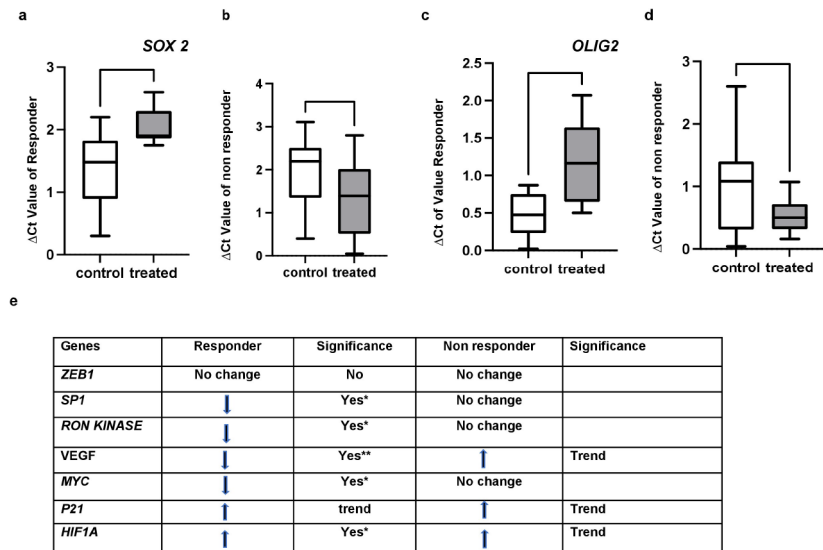


Figure 5. Cross-validation of Mit-A-responsive genes from *ex vivo* slice culture assay culture supernatant EVs. The slices were cultured with either Mit-A or vehicle control. Subsequently, supernatants were collected, and EVs were isolated for gene expression analysis via RT-qPCR. (a–d) These panels represent the expression levels of SOX2 and OLIG2 in responder and nonresponder populations by delta *Ct* values. ($\Delta Ct = Ct$ of target gene–*Ct* of reference gene; lower delta *Ct* values represents higher gene expression and vice-versa). (e) A tabular representation of Mit-A-responsive gene expression summarizing the findings. A *p*-value of less than 0.05 is considered statistically significant (**p* < 0.05; ***p* < 0.01; and ****p* < 0.001).

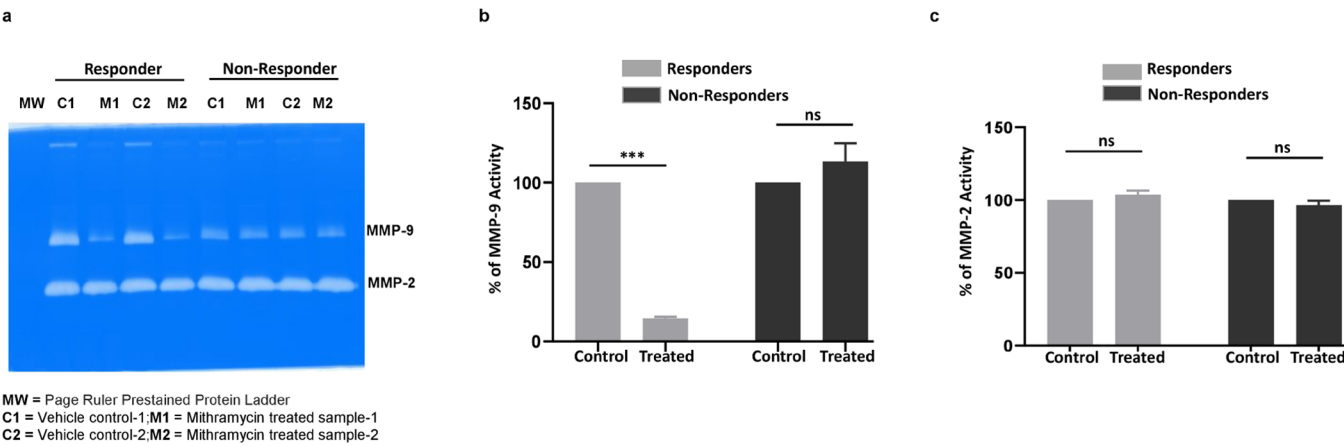


Figure 6. Analysis of MMP-9 and MMP-2 activity in *ex vivo* explant culture assay condition media. The analysis was conducted on conditioned media following treatment with Mit-A or vehicle control in the CANscript assay. (a) A representative image of gelatin zymography for MMP-9 and MMP-2 comparing Mit-A responder and nonresponder populations. (b,c) Quantification of the representative band intensities for MMP-9 and MMP-2 in both Mit-A responder and nonresponder groups. Vehicle control was set as 100% activity, and the respective activities of MMPs in the different populations were calculated and plotted. A *p*-value of less than 0.05 is considered significant (**p* < 0.05; ***p* < 0.01; and ****p* < 0.001).

gene cutoff values for all essential genes: SOX2 (2.096), OLIG2 (1.76), and ZEB1 (2.36). Utilizing these cutoff values, our findings demonstrated varying responses among the genes examined. SOX2 gene expression predicted 12 Mit-A responders and 33 nonresponders, OLIG2 gene expression predicted 25 Mit-A responders and 20 nonresponders, and ZEB1 gene expression predicted 27 responders and 18 nonresponders (Figure 4a–c). Furthermore, the stratified predicted responder population correlated with the tumor grade. Interestingly, the stratified Mit-A responders using the cutoff values of SOX2 and OLIG2 correlated with the glioma grade of the responder's group (grade IV) predicted by the *ex vivo* assay, whereas ZEB1-predicted responder population was a mixture of grades II, III, and IV. This outcome reinforces the potential value of this three-gene panel as a predictive tool for Mit-A response in glioma patients.

Mit-A Impact on Target Gene Expression in Culture Supernatants.

To elucidate the underlying mechanism of Mit-A efficacy among the responder groups, we measured the transcript levels of MYC, P21, HIF-1A, SP1, VEGF, SOX2, OLIG2, and ZEB1 in cell culture supernatant-derived EVs of the *ex vivo* chemosensitivity assay. Our findings showed significant downregulation of SOX2, OLIG2, VEGF, MYC, and SP1 expressions in the responder group compared to the nonresponders (Figure 5a–e). Conversely, we observed an upregulation of HIF-1A and P21 expression in the same group. To gain further insights, we conducted a protein–protein interaction (PPI) network analysis using the critical markers from this study. This analysis revealed a significant mechanistic connection (with a *p*-value confidence level of 5.09×10^{-13}) among proteins including OLIG2, SOX2, ZEB1, MYC, VEGF-A, MMP-2, MMP-9, and SP1 (Figure S3a,b). This gives a

detailed picture of the differential gene expression in response to Mit-A, potentially revealing the interconnected network of proteins that may be responsible for the observed therapeutic response.

Selective Inhibition of MMP-9 Activity by Mit-A in Glioma Responders. We determined the activities of MMP-2 and MMP-9 in culture supernatants obtained from the *ex vivo* assay by performing gelatin zymography. The resulting gelatin zymogram post Mit-A treatment for both the responder and nonresponder groups is depicted in Figure 6a. We quantified the band intensities corresponding to MMP-9 and MMP-2 using ImageJ software. The band intensity of the untreated media control was considered as 100% gelatinolytic activity for both MMP-2 and MMP-9 in the responder and nonresponder groups. The calculated densitometric values are presented in Figure 6b,c. Mit-A treatment significantly diminished MMP-9 activity in the responder group compared to the nonresponder group ($p < 0.003$, Figure 6a,b). However, MMP-2 activity remained unchanged in both the groups irrespective of Mit-A treatment (Figure 6a–c).

DISCUSSION

High-grade gliomas, particularly GBM, have long been associated with oncogenic drivers, such as EGFR, PDGFR α , and MET receptor tyrosine kinase. However, clinical trials targeting these gene mutations are yet to yield significant therapeutic effects.^{35,36} Recently, GSCs have emerged as the primary therapeutic targets for curbing the aggression of GBM.⁵ The possibility of repurposing established drugs like Mit-A would expand the landscape of potential therapies for GBM. Mit-A has shown potential for inhibiting the GSC transcriptional network, particularly the SOX2, OLIG2, and ZEB1 genes that are critical drivers of GBM progression, making it a novel repurposed drug for targeting gliomas.⁷ Here, we reported the therapeutic response of gliomas to Mit-A through an *ex vivo* assay and subsequently developed a Mit-A-responsive diagnostic gene panel (SOX2, OLIG2, and ZEB1) that could predict Mit-A sensitivity by quantifying mRNA expression levels in serum-derived EVs.

In recent years, circulating serum biomarkers have been beneficial in predicting therapeutic responses and providing minimally invasive alternatives to biopsies.³⁴ Moreover, the practical application of serum EV cargo is hindered by technical limitations inherent in current EV isolation methods as well as variations in EV content within the tumor microenvironment across diverse patient profiles. We focused on the EV compartment to assess the glioma tumor response to drugs. We confirmed the purity of the isolated glioma serum EVs by size, shape, and signature surface markers. Serum EVs are spherical, double-membrane, and intact structures ranging from 30 to 200 nm. Tumor-specific serum EV RNA has shown promise in identifying pathological and diagnostic features in several types of cancers, including gliomas.^{34,37–39} Thus, we utilized RT-qPCR to quantify GSC transcription marker gene expression in serum EVs.

Our research validates the suppressive action of Mit-A on the proliferation of GBM. The assessment hinges on the results of an *ex vivo* chemosensitivity assay, which measures alterations in tumor morphology, cell viability through hematoxylin and eosin (H&E) staining, proliferative activity via Ki67 immunostaining, and apoptosis induction as evidenced by cleaved Caspase-3 staining. These observations are congruent with findings reported in the literature.^{20–22} In

the course of our investigation, we discerned two distinct groups of glioma patients differentiated by their response to Mit-A. This bifurcation was facilitated by a proprietary algorithm that computes an M-score, incorporating data from the functional assays described above. Patients were categorized into Mit-A responders with an M-score greater than 25 and Mit-A nonresponders with an M-score less than 25. This stratification underscores the heterogeneity of glioma patient responses to Mit-A and presents a quantitative basis for assessing the efficacy of Mit-A in individual patients. Interestingly, all Mit-A-responsive patients were observed orienting toward high-grade glioblastomas; this aligns with recent findings which reported that Mit-A inhibits GBM invasiveness in patient-derived high-grade gliomas.^{21,22} Moreover, the IDH1 mutation status of the tumor significantly complemented the M-score, suggesting a positive correlation between wild-type IDH1 status and response in the *ex vivo* setting. This finding contrasts with previous reports that posited Mit-A as a potent inhibitor for IDH mutant high-grade gliomas.²¹ This discrepancy could be explained by differences in the sample types used in the respective studies.

Our investigation sought to illuminate the molecular underpinnings of differential responsiveness to Mit-A by examining a spectrum of downstream signaling molecules, including MYC, VEGF, P21, OLIG2, ZEB1, and SOX2. We also evaluated the activity of matrix metalloproteinases MMP-2 and MMP-9 due to their established association with glioma progression and poor patient outcomes.⁴⁰ These molecular entities were quantified in EVs obtained from cell culture supernatants in the *ex vivo* assay. Upon treatment with Mit-A, the responder cohort demonstrated downregulation of SP1, SOX2, OLIG2, MYC, VEGF, and RON, coupled with an upregulation of P21 and HIF-1A and a concomitant decrease in MMP-9 activity, which is in line with previous studies.^{7,20,41} The findings suggest that Mit-A's antitumor mechanism may involve modulating key genes and protein expression related to cancer cell proliferation, angiogenesis, and extracellular matrix remodeling. However, MMP-9 also exhibits antitumor activity and plays vital role in physiological functions. To assess the potential therapeutic efficacy of inhibiting MMP-9 function in Mit-A responders, it is imperative to develop specific MMP-9 inhibitors that selectively target the tumor-promoting function while preserving its antitumor effects. Unfortunately, as of now, none of the MMP-9 inhibitors have received FDA approval.⁴²

The selective antitumor efficacy of Mit-A observed in our study may be attributed to the cellular heterogeneity inherent in brain tumors. This heterogeneity is potentially rooted in the presence of GSCs, which are capable of initiating distinct tumor cell lineages post tumor initiation.⁵ GSCs are pivotal in mediating resistance to current therapies and are implicated in the recurrence of GBM, thus representing a critical focus for understanding treatment response and improving therapeutic outcomes.²

Recent research has illuminated the significance of a cohort of transcription factors, including SOX2, OLIG2, Zeb1, SALL2, and POU3F2, which are notably overexpressed in GBM.^{7,43} Evidence suggests that the exogenous expression of these transcription factors can reinstate a glioma stem cell-like phenotype, pointing to their critical role in maintaining the malignancy of GBM cells.⁷ Further, investigations by Singh et al. and Alonso et al. have highlighted SOX2 overexpression in high-grade gliomas, a phenomenon that is rooted in epigenetic alterations such as hypomethylation of the SOX2 promoter's

CpG island and increased acetylation of H3K27 histone, underscoring a complex regulatory modification pattern.^{7,44}

Building upon these observations, our study posits that the discerning antitumor impact of Mit-A on high-grade gliomas may be mediated through its interaction with these epigenetic alterations, thereby influencing the expression of pivotal genes and proteins. Delving into the epigenetic landscapes of Mit-A-responsive and nonresponsive glioma cases could shed light on the selective antitumor effects we have observed and unravel the epigenetic intricacies that differentiate the treatment responses in GBM patients. Such an in-depth understanding could pave the way for targeted therapies that exploit these epigenetic vulnerabilities in glioma. The PPI network analysis provides further insights into the potential molecular pathways and interactions contributing to Mit-A's therapeutic effects in glioblastoma.

The functional response profile generated from the *ex vivo* tumor sensitivity assay was translated into a molecular assay that collectively reflected the Mit-A sensitivity. We established and validated an EV-based assay for detecting SOX2 and its associated genes OLIG2 and ZEB1 by RT-qPCR method. All responder tumors in the population have increased SOX2 and OLIG2 and an increasing trend of ZEB1 expression levels. The diagnostic potential to identify the predictive power of SOX2, OLIG2, and ZEB1 was assessed using the area under ROC curve analysis. SOX2 and OLIG2 were found to be highly predictive as potential biomarkers of Mit-A response. The diagnostic potential of SOX2 and OLIG2 in relation to Mit-A was high, with predictive powers of 80 and 79%, respectively. Both showed outstanding sensitivity and specificity. Overall the combined gene panel (SOX2, OLIG2, and ZEB1) yielded a high predictive power (95.6%) with a sensitivity of (94.1%) and a specificity of (100%), thus suggesting their high utility as noninvasive diagnostic biomarkers for Mit-A response. SOX2, OLIG2, and ZEB1 biomarkers independently demonstrated a robust PPV but limited NPV in predicting Mit-A therapy outcomes for glioma patients. Integration of these markers enhances overall precision, effectively distinguishing responders and nonresponders, and thereby refining the personalization of Mit-A treatment for optimal efficacy.

This study has discerned a pivotal distinction between Mit-A responders and nonresponders by applying a three-gene panel, marking a significant advancement. The direct clinical implications of employing this gene panel for immediate therapeutic use necessitate additional validation across a broader glioma patient population. The application of our findings has the potential to inform the creation of custom-tailored treatments for glioma, particularly through the innovation of less toxic Mit-A derivatives and advanced drug-delivery modalities, such as exosomal and liposomal vectors. These advancements are promising for augmenting the therapeutic efficacy of Mit-A while minimizing its systemic adverse effects and enhancing its ability to penetrate the brain and target tumors effectively.

Our study underscores the therapeutic promise of Mit-A as an adjunct or alternative to current standard treatments for GBM. Given that Mit-A has already secured FDA approval, we foresee that our results could catalyze more comprehensive clinical trials and potentially broaden the therapeutic applications of Mit-A for patients with high-grade GBM. Furthermore, assessing the expression levels of our gene panel (SOX2, OLIG2, and ZEB1) in glioma patients facilitates a precise classification into Mit-A-responsive or nonresponsive

groups. This stratification is anticipated to curtail adverse reactions associated with Mit-A, refine therapeutic strategies, and ultimately expected to markedly improve clinical outcomes for patients afflicted with glioma.

While our study reveals promising findings on the potential use of Mit-A for high-grade gliomas, it bears several limitations that must be considered. Our investigation was confined to limited patient samples for establishing tumor sensitivity testing. Expanding the sample size in subsequent studies would strengthen the applicability of the results to diverse patient populations. The unavailability of IDH1 mutation status for all samples in our study poses another constraint, considering IDH1 mutations' pivotal role in glioma prognosis and classification. For a more profound understanding of the influence of IDH1 mutations on the efficacy of Mit-A, future research endeavors must encompass detailed data on these mutations. Moreover, an extensive analysis of the transcriptomic and epigenetic landscapes of Mit-A responders and nonresponders is crucial. Such studies will be instrumental in unraveling the molecular underpinnings that dictate the sensitivity to Mit-A in glioma. By identifying the specific genes and pathways implicated, these investigations will not only elucidate the mechanisms through which Mit-A exerts its effects but also potentially pave the way for more targeted and effective treatment modalities. Given these limitations, our study should be considered a promising starting point, warranting further comprehensive research to validate our findings and bolster our understanding of the Mit-A role in treating high-grade gliomas.

■ CONCLUSIONS

Our study employed an *ex vivo* tumor sensitivity assay to establish Mit-A as a repurposed drug for treating GBM. GSCs can be identified and are quantifiable through the SOX2 transcriptional target gene panels OLIG2 and ZEB1 in serum EVs. Finally, the EV-derived mRNA transcript assay gene panel predicts Mit-A sensitivity, offering a potential biomarker for selecting optimal Mit-A regimens in high-grade gliomas. Altogether, we report a proof-of-concept for assessing Mit-A responsiveness in GBM patients. This study provides critical translational and mechanistic insights for conducting clinical trials with Mit-A and its new analogues. It will pave the way for the biomarker-guided optimal selection of Mit-A regimens for high-grade gliomas.

■ METHODS

Patient Sample Collection. Tissue samples from tumors and corresponding blood specimens were collected from a cohort of 70 glioma patients following surgical procedures, with the process being conducted in strict adherence to ethical guidelines, including informed consent and approval by the Apollo Hospitals Ethics Committee (reference: AHJ-ACD-071/08-21). The histological assessment of the acquired tumor specimens was carried out in alignment with the ISNO consensus guidelines, effectively integrating the practical aspects of the WHO 2016 criteria for classifying adult diffuse gliomas.⁴⁵ Our molecular diagnostic approach was underpinned by the inclusion of three pivotal biomarkers: the 1p19q codeletion, mutations within the isocitrate dehydrogenase genes (IDH1 and IDH2), and the alpha thalassemia/mental retardation syndrome X-linked (ATRX) gene alterations. The evaluation of these biomarkers was instrumental for the

stratification of glioblastomas into primary or secondary categories, informed by the IDH1 mutational status alongside other molecular indicators. Our investigation spanned gliomas of varying histological grades, ranging from grade-I to grade-IV and included cohorts with both wild-type and mutant forms of IDH1. Blood was collected from age- and gender-matched 10 healthy volunteers from the blood bank. Following surgical removal, all tissue samples were promptly flash-frozen in “RNAlater” (Life Technologies, San Francisco, California, USA) to preserve RNA integrity for downstream analyses. For ex vivo live tumor slice cultures, tissues were stored in Aquix buffer at a consistent temperature of 4 °C. Healthy brain tissues from three individuals were sourced from the NIMHANS brain biobank.

Patient-Derived Ex Vivo Culture of Tumor Slices. The CANScripTTM assay, a patient-derived ex vivo drug sensitivity test developed at Farcast Biosciences (formerly known as Mitra Biotech Pvt Ltd., Bangalore, India) was utilized to evaluate the sensitivity of glioma tumors to Mit-A. Briefly, tumor tissues were sectioned into 2–3 mm³ slices. These slices were subsequently placed in a carefully designed individualized microenvironment for culture in triplicates as described previously.⁴⁶ Mit-A treatment was initiated with the recreated tumor microenvironment at a dose of 100 nM along with vehicle control (DMSO) for 72 h. Post treatment, the response of the tumor slices to Mit-A was evaluated by assessing changes in morphology, viability, and cell death in 10% formalin fixed paraffin embedded cultures through histopathological examination. The data obtained from these measurements were fed into a previously described algorithm to generate an M-score.²⁶ This score is predictive of the clinical response to Mit-A treatment. An M-score greater than 25 indicates that the patient’s tumor tissue is likely to respond positively to Mit-A, and a score below 25 predicts its nonresponsiveness to the drug.

Tumor Morphology and Immunohistochemistry. The tumor slices were processed, fixed in formalin, and embedded in paraffin after 72 h of culturing. Tumor sections, 5 μm thick, were stained with H&E to examine their morphology, viability, and necrosis. FFPE tumor sections were also antigen retrieved (using Vector Lab), blocked with normal goat serum (5% in PBS, Vector Lab), and then stained with primary antibodies such as Ki-67 (mouse monoclonal, Dako, MIB1, M7240) and Caspase 3c (rabbit polyclonal, Cell Signaling Technology, D-175) to examine the proliferation and apoptosis induction, respectively. Secondary antibodies such as HRP-conjugated goat antimouse IgG (Envision Kit, Dako, K4001) or goat antirabbit IgG (Signal Stain Boost IHC detection reagents, Cell Signaling Technology 81145) were used for Ki67 and Casp3c, respectively. Finally, chromogenic DAB detection (Vector Lab) was used for the final detection of stained slides. Hematoxylin was used to counterstain the stained slides. Moreover, IDH1R132H and ATRX proteins were immunostained using the Ventana BenchMark XT immunostainer (Ventana Medical Systems, Tucson, USA) as per the manufacturer’s instructions. The antibodies used for immunostaining were IDH1 R132H (H09, Dianova, Hamburg, Germany, dilution 1:40) and ATRX (HPA001906, Sigma, dilution 1:300). The positive and negative controls were run alongside our samples. The stained slides were imaged under a bright field at 200X magnification using a light microscope (Leica DM2500).

EV Isolation. The clarified serum samples were obtained by centrifugation at 2000g for 20 min at room temperature. The total EV isolation kit (Thermo Fisher Scientific, USA) precipitated EVs from the serum and the tumor supernatant. The process of sample collection and preservation from glioma patient explant cultures post treatment was conducted with high precision. Conditioned media were carefully harvested and immediately stored at –80 °C to maintain the integrity of the samples for further analysis. Upon collection, the supernatants were subjected to centrifugation at 5000g for a duration of 10 min, a critical step designed to remove any cellular debris. Thereafter, EVs were meticulously separated from the tumor-derived conditioned media using the total EV isolation kit, Thermo Fisher Scientific, USA. This methodological approach ensures the reproducibility and reliability of the study, facilitating the accurate assessment of the EV content and function.

EV Characterization. EV quality and quantity were established at the biochemical, biophysical, and molecular levels.

Western Blot Analysis. The detection of EV markers (Annexin-5, CD54, Flotillin-1, and EPCAM) was evaluated using Western blot analysis. Thirty μg of serum EV protein lysate was briefly loaded onto 10% SDS-PAGE gels. The protein was transferred onto poly(vinylidene fluoride) membranes (EMD Millipore, Billerica, CA, USA). The EV markers were determined using the primary antibodies (my BioSource, USA) at a dilution of 1: 1000 for 3 h at room temperature (RT). The primary antibodies were washed in 0.05% Tween-20/PBS and then incubated with a goat antimouse and antirabbit horseradish peroxidase-conjugated IgG secondary antibody (cat no: MBS2549984, my BioSource, USA) at a dilution of 1: 2500 dilution for 60 min at RT. The signal was visualized by adding an enhanced chemiluminescence reagent (Bio-Rad Laboratories, Inc., USA). Images were captured by using a Chemi Doc MP imaging system (Bio-Rad Laboratories, Inc., USA).

TEM. To confirm their morphology, the isolated serum EVs were analyzed using TEM (JEM-2100, JEOL Ltd.) at the Centre for Cellular and Molecular Biology (CCMB), Hyderabad. In brief, EV samples were fixed with 1% glutaraldehyde for 5 min on 400-mesh copper grids (FCF400-Cu, Electron Microscopy Sciences). Then, the TEM grid was stained with 2% uranyl acetate and air-dried at room temperature. Images of the isolated serum EVs were captured for further analysis.

EV Size Analysis. EV concentration and size distribution were determined using NTA. The serum EV samples were analyzed with the Nano Sight LM10 instrument (Nano Sight, Amesbury, UK). The LM10 uses a combination of laser light microscopy and Brownian motion to determine EVs’ size, concentration, and morphology. To improve separation, the EVs were diluted with particle-free PBS (0.1 μm filtered) in the ratio of 1:100, resulting in a particle concentration within the recommended range of 10⁸–10⁹ particles/mL.

Semiquantitative and Quantitative RT-PCR. According to manufacturer instructions, RNA was extracted from the tumor tissue and EVs using a Trizol and Trizol LS reagent (Thermo Fisher Scientific, USA). RNA quantification was done with a Qubit RNA BR assay kit (ThermoFisher Scientific, USA). Isolated EV RNA with RIN values of 9–10 were measured by the Agilent Bioanalyzer 2.1 instrument using the RNA Pico (Agilent, USA). One microgram of total EV-derived

RNA was reverse transcribed with a high-capacity cDNA reverse transcription kit (Invitrogen, USA). The cDNA was preamplified using Sapphire Amp fast PCR master mix (Takara Bio, USA). The primer sequences used to amplify Mit-A target genes are listed in Table S1. PCR cycle includes denaturation at 95 °C for 5 min followed by 20 cycles with denaturation at 95 °C for 1 min, annealing at 60 °C for 30 s, extension at 68 °C for 1 min, and a final extension at 72 °C for 5 min. One μ L of preamplified PCR (1:10 dilutions) was used as a template in the PCR reaction using the same primer sets and cycling conditions. PCR was performed on the Master Cycler pro-S model (Eppendorf, USA). PCR products were analyzed on 2% agarose gel using 5 μ L of the PCR product; the gel was stained with ethidium bromide and visualized on the Bio Spectrum 600 imaging system (UVP, U.K.).

For RT-qPCR, 1 microliter of preamplified PCR (1:10 dilutions) was used as a template in the PCR reaction containing each primer at a concentration of 0.1 μ M. In brief, a 10 μ L PCR reaction was performed in an Applied Biosystem 7500 Real-Time System using 5 μ L of SYBR Premix Ex TaqII ROX plus (Takara Bio USA, Inc.) and 0.15 μ L of 10 μ M primer pair (Bio serve). The cycling conditions of RT-qPCR were the same for all genes (*SOX2*, *OLIG2*, *ZEB1*, *c-myc*, *p21*, *SP1*, *HIF-1 α* , *VEGF*, and *GAPDH*). The initial denaturation step was at 30 s at 95 °C, followed by 30 cycles of denaturation at 95 °C for 5 s and annealing step at 60 °C for 30 s. All samples were evaluated in triplicates and ran with nontemplate controls. For all samples, the validity of the RT-qPCR experiments relied on the validation of both positive and negative controls. The expression levels of all genes were normalized to that of GAPDH and were presented as delta *Ct* [$\Delta Ct = Ct$ (gene of interest) – *Ct* (housekeeping gene)]. These delta *Ct* values exhibit an inverse correlation with the gene expression level. Further, these delta *Ct* values were plotted. For each patient among Mit-A responders and nonresponder groups and subsequently utilized for statistical analyses such as Student's *t* test and ROC curve analysis.

Zymography. Gelatin zymography was used to analyze MMP-9 and MMP-2 activities in the tumor condition media from the ex vivo chemosensitivity assay.⁴⁷ Briefly, PAGE (8%) containing 0.2% gelatin was loaded with 5 μ L of each Mit-A control and treated tumor CANScripT-derived condition media. PAGE was carried out at 80 V for 1 h 30 min until the dye front reached the gel's bottom. The gels were washed with 2.5% triton X for 40 min, incubated with incubation buffer at 37 °C for 20 h, and stained with Coomassie brilliant blue for 1 hr. The gels were then destained with the destaining solution until clear gelatinolytic activity bands were in the gel's blue background. The gels were imaged using a Gel Doc XR+ Gel documentation system (Bio-Rad Laboratories, Inc., USA).

PPI Analysis. The study utilized the STRING network analysis (<https://string-db.org>) to investigate PPIs. Input markers, including *SOX2*, *OLIG2*, *ZEB1*, *SP1*, *HIF-1A*, *Myc*, *P21*, *MMP-9*, *MMP-2*, and *VEGF*, were queried at a high confidence level of 0.700 for network visualization. After the PPIs were searched, a network was constructed on the STRING website.

Statistical Analysis. Statistical analyses were conducted using SPSS version 24.0 (SPSS Inc., Chicago, USA) and GraphPad Prism 9.4 (GraphPad Software, Inc., San Diego, California, USA). Descriptive statistics were calculated for all variables. Quantitative variables were expressed as the mean $\pm$ standard deviation, while categorical variables were described

using frequencies and percentages. Statistical associations between variables were explored using Student's *t* tests for independent samples, chi-square tests for categorical variables, and Mann–Whitney U-tests for nonparametric data. The diagnostic performance of serum EV biomarkers (*SOX2*, *OLIG2*, and *ZEB1*) in glioma patients was assessed through ROC curve analysis. The respective AUC was calculated to evaluate the predictive ability of these EV-derived mRNA biomarkers in differentiating between Mit-A responders and nonresponders. To determine the optimal cutoff value for each mRNA expression level, we identified the point providing the maximum sensitivity and specificity in distinguishing between Mit-A responder and nonresponder groups, with their 95% CIs. Univariate logistic regression analyses were employed to assess the associations between various risk factors and the presence of glioma. A *p*-value of less than 0.05 was considered statistically significant in all tests.

■ ASSOCIATED CONTENT

Data Availability Statement

The data supporting the conclusion of this article are included in the report. Any request for data may be sent to the corresponding author.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acspsci.3c00198>.

Additional information about primers used in the study (Table S1), establishment of GSC transcription factors *SOX2*, *OLIG2*, and *ZEB1* in the glioma tissue and EVs (Figure S1), expression profiles of *SOX2*, *OLIG2*, and *ZEB1* in healthy individuals and glioma patients (Figure S2), and PPI network analysis using the critical markers from this study (Figure S3) (PDF)

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Notes

This study was initiated after approval by Apollo Hospitals Ethics Committee—Biomedical Research (IEC-BMR), Apollo Hospitals IEC Application no AHJ-057/05-17 with a study protocol no. CMBRC/2017/002.

The authors declare the following competing financial interest(s): Biswanath Majumder, Dilli Kumar, Manjusha Biswas, Pradip K. Majumder, and Ajith V. Kamath were previous employees of Farcast Bioscience; Dr. Sasidhar, Director of Urvogel Bio pvt Ltd; and Dr Amit, CEO of Exsegen Research company. The remaining authors declare no competing financial interests concerning the study.

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